



Scope of work and structured thinking

Earlier I told you that carefully defining a business problem can ultimately save time, money, and resources. All of this is achieved through structured thinking. **Structured thinking is the process of recognizing the current problem or situation, organizing available information, revealing gaps and opportunities, and identifying the options.** In other words, it's a way of being super prepared. It's having a clear list of what you are expected to deliver, a timeline for major tasks and activities, and checkpoints so the team knows you're making progress. In this video, we'll look at how structured thinking helps us save time and effort, but also makes our job as data analysts easier because it allows us to better understand the work we are doing. In the business world, it's common for teams to spend hours of valuable time trying to solve an important problem, only to end up back where they started. Not only is the initial problem not resolved, but they've spent hours not resolving it. This outcome negatively affects you, your team, and the organization as a whole. But it can usually be prevented. Many times the situation is a result of not fully understanding the issue. **Structured thinking will help you understand problems at a high level** so that you can identify areas that need deeper investigation and understanding. **The starting place for structured thinking is the problem domain**, which you might have remembered from earlier. Once you know the specific area of analysis, you can set your base and lay out all your requirements and hypotheses before you start investigating. With a solid base in place, you'll be ready to deal with any obstacles that come up. What kind of obstacles? Well, let's say you're asked to predict the future value of an apartment building based on a given dataset. You have hundreds of variables and every one is crucial to your analysis. But what if one variable accidentally gets left out, like square footage, for example? You'd have to go back and redo all your hard work. That's because missing variables can lead to inaccurate conclusions. Another way that you can practice structured thinking and avoid mistakes is by using a **scope of work**. **A scope of work or SOW is an agreed-upon outline of the work you're going to perform on a project.** For many businesses, this includes things like work details, schedules, and reports that the client can expect. Now, as a data analyst, your scope of work will be a bit more technical and include those basic items we just mentioned, but you'll also focus on things like **data preparation, validation, analysis of quantitative and qualitative datasets, initial results**, and maybe even some visuals to really get the point across. Let's bring a scope of work to life with a

simple example. Say a couple has hired a **wedding planner**. We'll focus on just one task, the **wedding invitations**. Here's what might be in **scope of work**: **deliverables**, **timeline**, **milestones**, and **reports**. Let's break down just one of these, **deliverables**. The wedding planner and couple will need to decide on the invitation, make a list of people to invite, collect their addresses, print the invitations, address the envelopes, stamp them, and mail them out. Now let's check out the timelines. You'll notice the dates and the milestones which keep us on track. Finally, we have the reports, which give our couple some peace of mind by telling them when each step is complete. A scope of work can be a simple but powerful tool. With a solid scope of work, you'll be able to address any confusion, contradictions, or questions about the data **up- front** and make sure these sneaky setbacks don't stand in your way. This is a simple example of what a scope of work might look like. But later, you'll be able to practice building your own. Next up in our scope, we'll check out setbacks from a different angle by learning the importance of **contextualizing** data and avoiding bias. Looking forward to sharing some cool insights with you.

Reflect

Be sure to differentiate statement of work from scope of work, which are both abbreviated as SOW. Although they help define deliverables and a timeline, they aren't the same and shouldn't be used interchangeably.

A statement of work is a document that clearly identifies the products and services a vendor or contractor will provide to an organization. It includes objectives, guidelines, deliverables, schedule, and costs.

A scope of work is project-based and sets the expectations and boundaries of a project. A scope of work may be included in a statement of work to help define project outcomes.

As a junior data analyst, it's more typical to be asked to create a scope of work than a statement of work.